

MuZero-Mario: what works, what fails, and how to resume

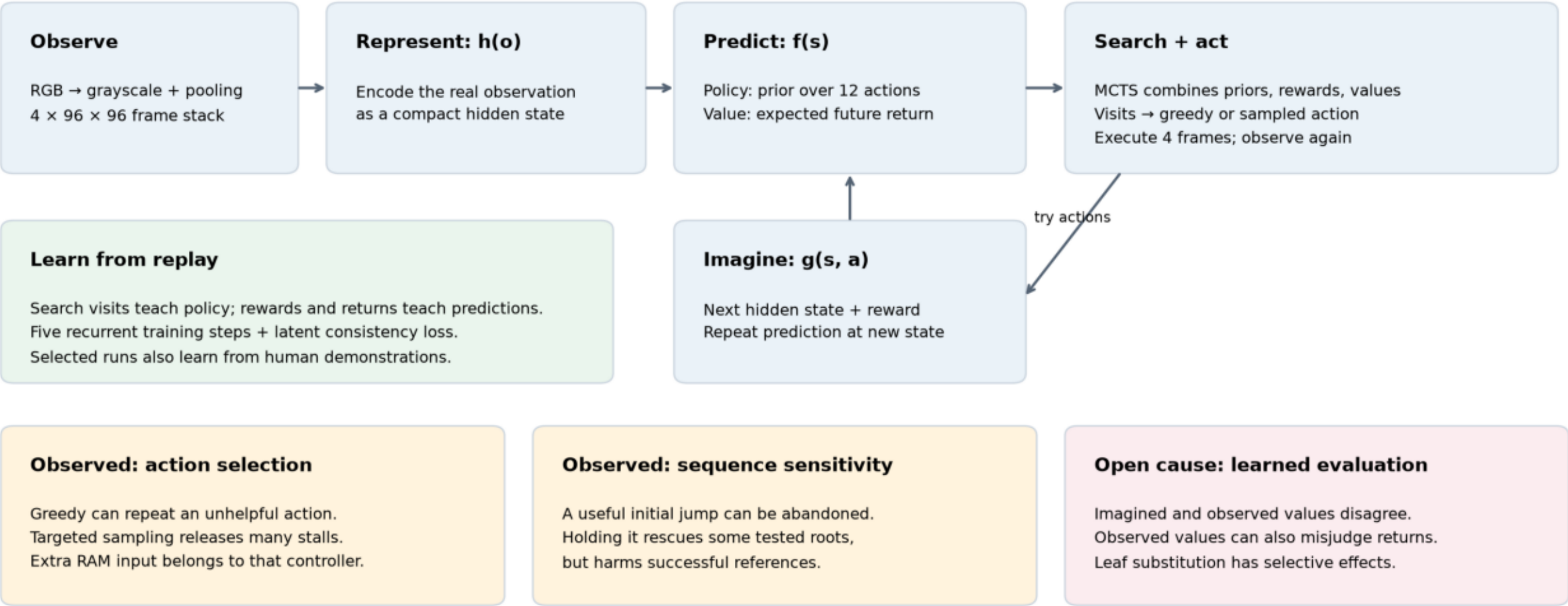
- A research handoff • 20 September 2026
- Partial specialist competence; broad human-level performance is not established.
- A replicated controller improvement and controlled tests now narrow the failure mechanisms.

What would human-level performance mean?

- Reliability: completing a level on the first life, counting deaths and timeouts.
- Speed: time among successful attempts, reported separately.
- Breadth: all 22 human-covered levels; 21 currently evaluated model levels, with 5-3 missing.
- Robustness: starts and training seeds; a specialist fleet is not a single generalist.

Inside MuZero: observe, imagine, search, act

What MuZero does at each decision — and where failures can arise



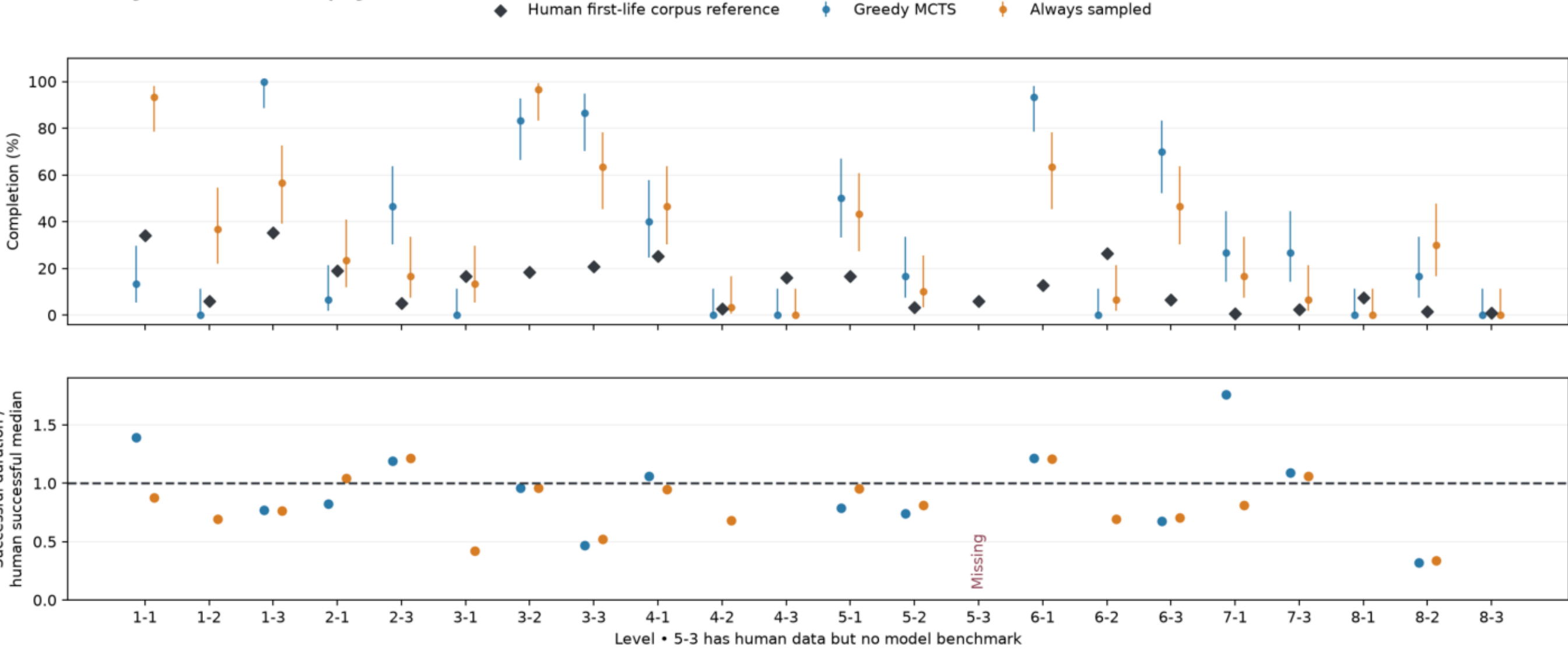
Conceptual map, not a measured attribution diagram. MuZero predicts task-relevant latent states, not future screenshots. The benchmark uses 50 simulations with leaf batch 4. The network is reused after every real action; search does not commit to a whole plan. Read docs/handoff/REPORT.md for evidence strength, human comparison limits and the next discriminating experiments.

What the project has built

- Specialists across the human-covered corpus; archived model bundles include training provenance.
- Human demonstrations rescue some self-play failures; multi-level curriculum arms and a distillation pipeline exist.
- Frozen checkpoint evaluations, exact replay and paired interventions separate several failure types.
- A portable report, reproducible figures, checkpoint examples and an experiment/data registry.

Where we stand against recorded humans

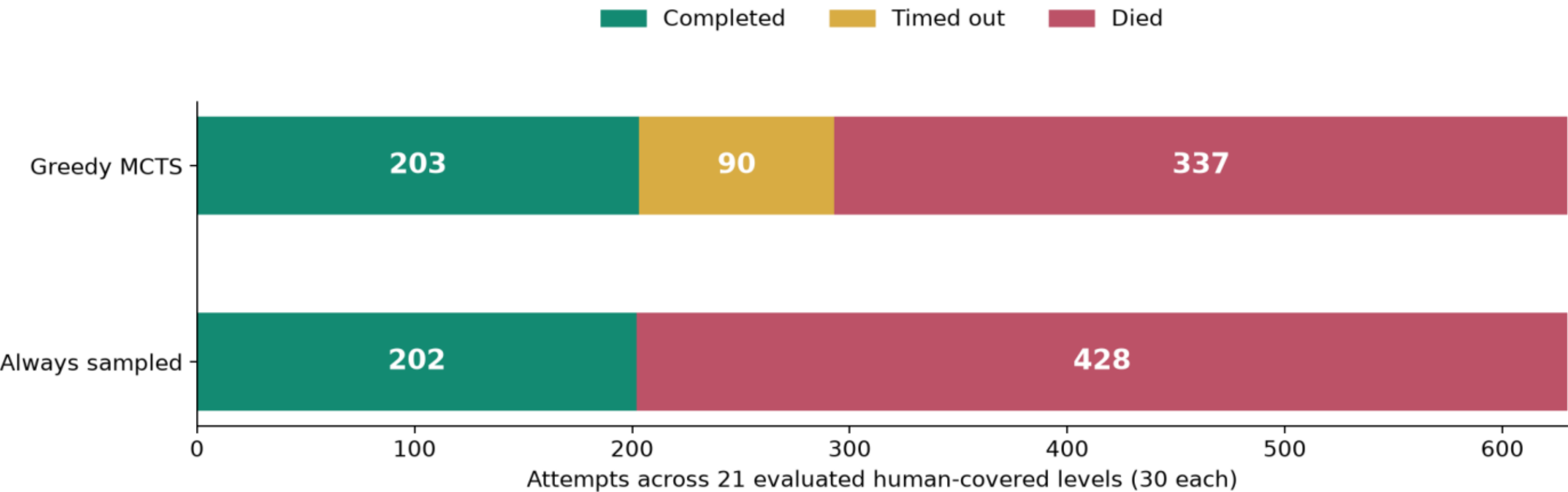
Reliability varies sharply across the human-covered levels



30 development attempts per model/controller; model error bars are fixed-checkpoint Wilson 95% intervals. Human rates pool repeated attempts from five people. Speed dots condition on success; human successful lives include respawns. Starts, practice and budgets differ. Faster successful runs do not establish human-level reliability.

Removing stalls is not enough

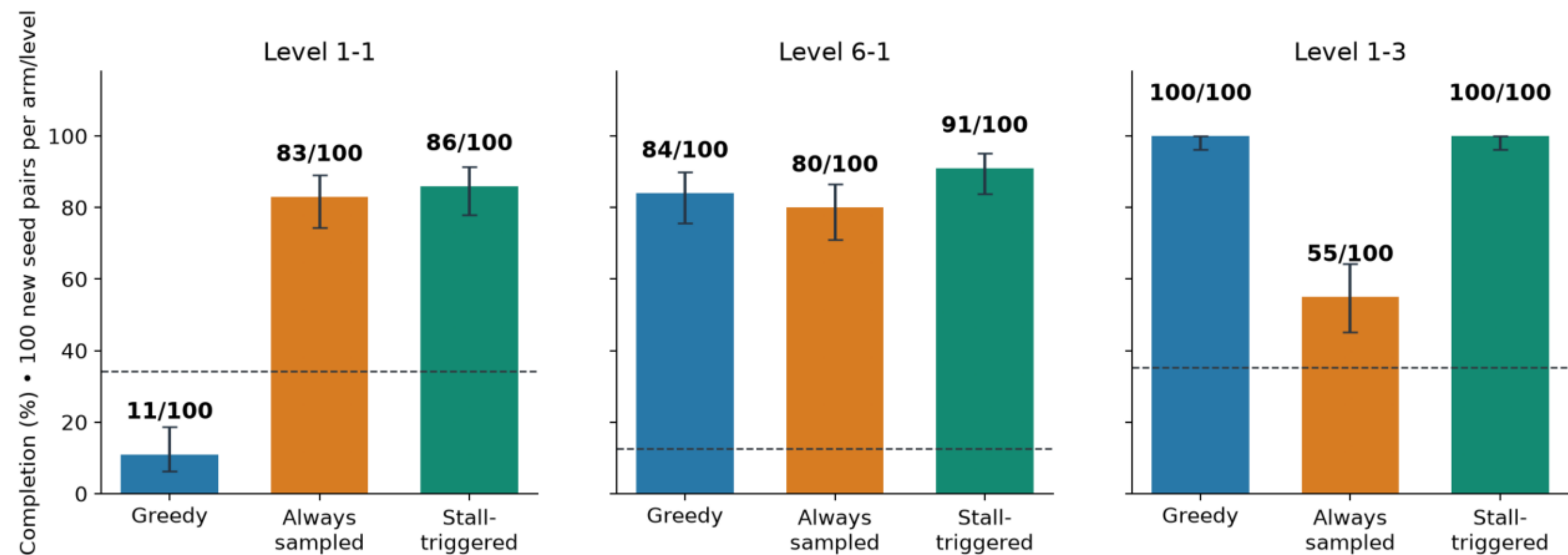
Always-on sampling removes stalls, but increases deaths



Development evidence, fixed checkpoints: 203/630 (32.2%) greedy versus 202/630 (32.1%) sampled completions.
Equal attempts per level; no claim of universal controller superiority. Level5-3 is missing; 2-2/7-2 are excluded for absent human data.
These totals intentionally differ from the historical 22-model-level report, which included 2-2.

A practical improvement that replicated

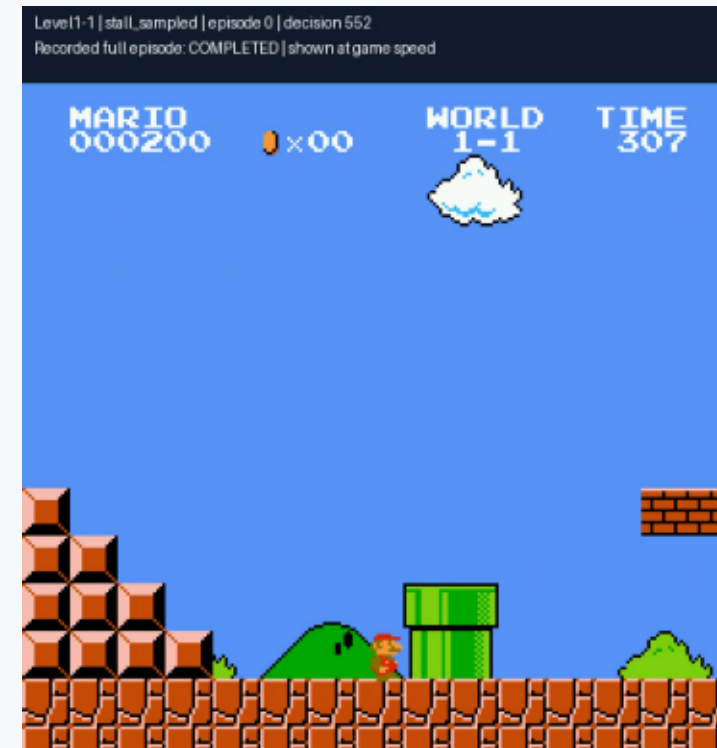
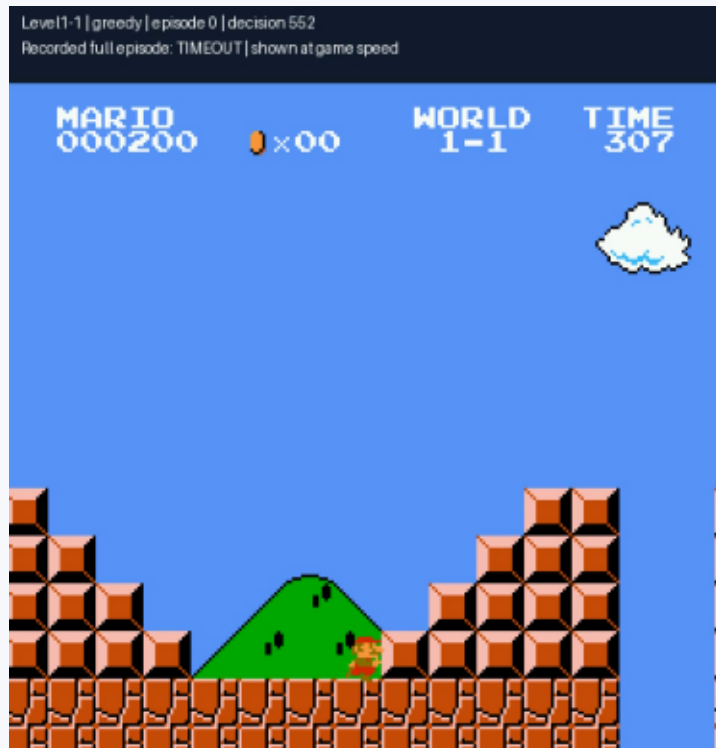
Intervene at stalls: a confirmed controller improvement on three levels



900 completed episodes; new confirmation seeds, unchanged checkpoints, 50 simulations and no root noise. Error bars: Wilson 95%.
Dashed line: descriptive human first-life corpus rate (protocols differ). Stall trigger uses extra RAM x/player-state input.
On 1-1, gated sampling rescues 76/78 greedy timeouts but loses one greedy success. Evidence does not establish 86% > 83% or all-level benefit.

Watch the same recorded start under two controllers

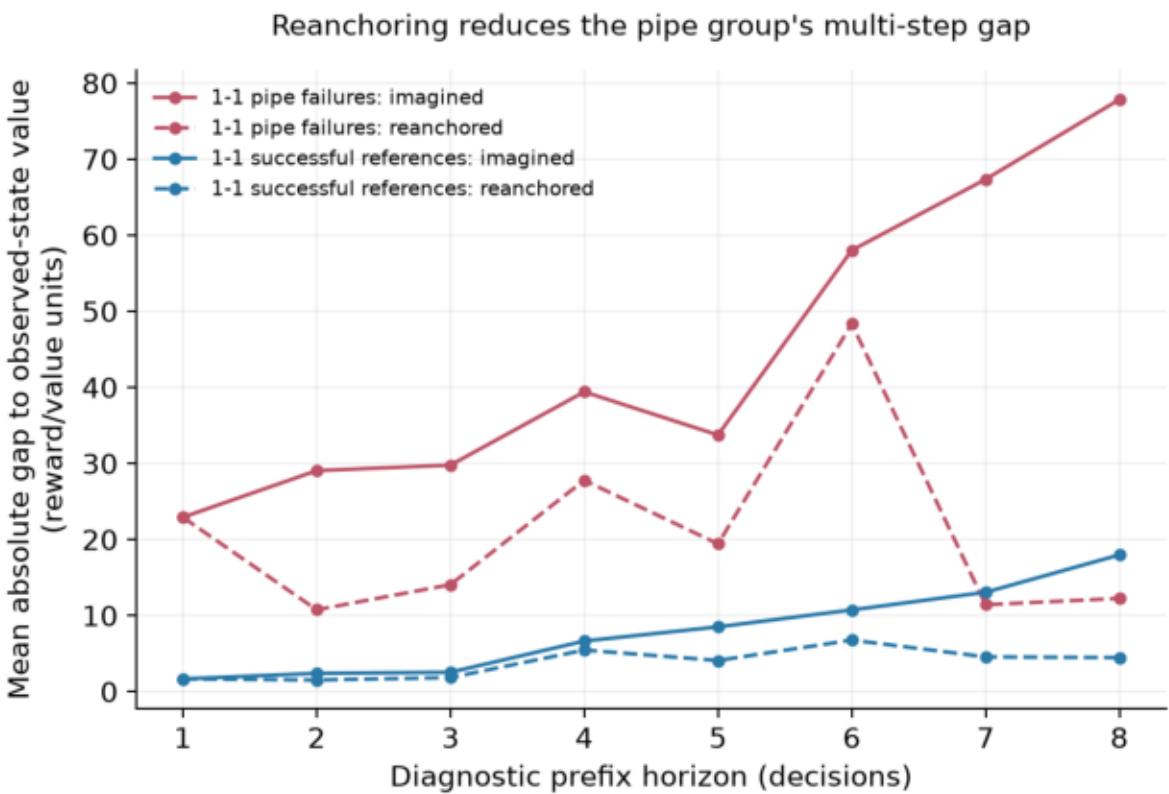
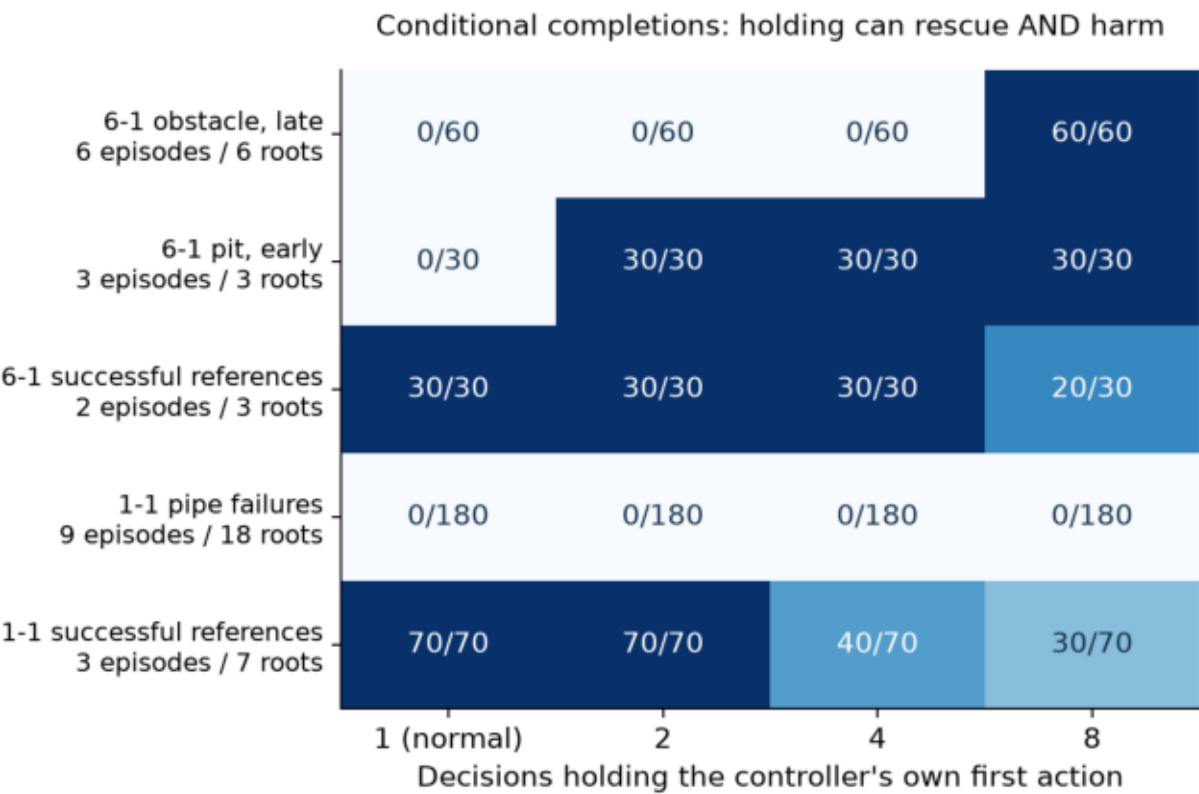
- Matched confirmation episode; first qualifying timeout-to-success pair by episode index.
- Clips show a fixed window around intervention, at emulated game speed. Full trajectories were replay-verified.



Static preview: play the paired videos in the HTML deck for the time course.

Action timing can help—and harm

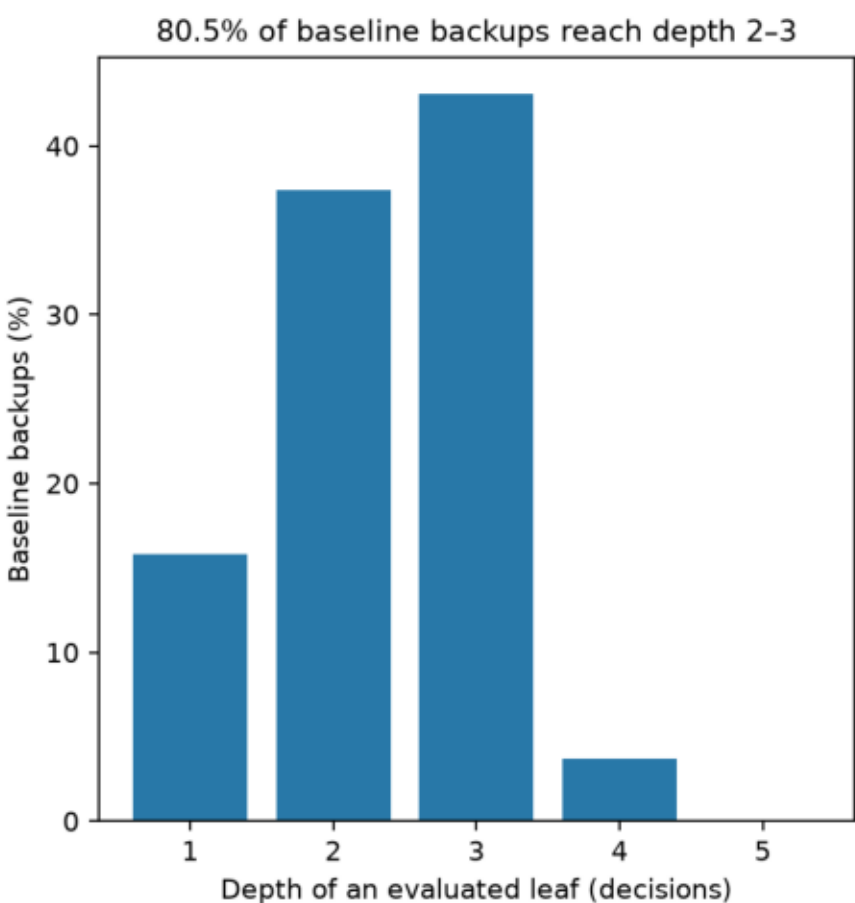
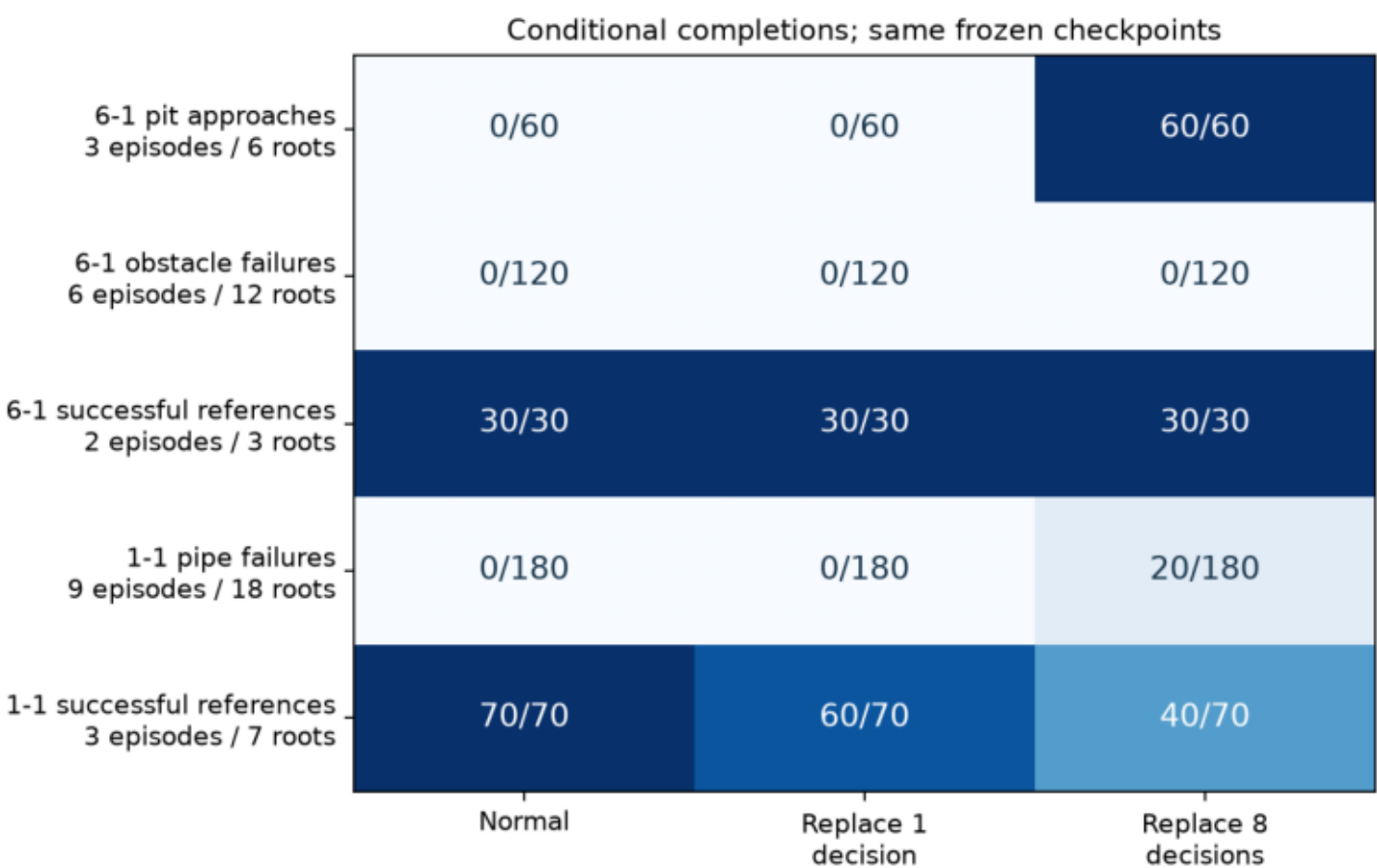
Two distinct clues: action timing and imagined-value inconsistency



Left: selected greedy-source groups; 10 conditional seeds/root, not independent level attempts. 1 decision = 4 emulator frames.
Right: equally weighted roots x four diagnostic prefixes; 18 pipe roots from 9 episodes, 7 reference roots from 3 episodes. No population CIs.
Observed-state values come from the same network, not ground truth. Fixed prefixes need not be paths explored by MCTS.
These results do not isolate dynamics, BatchNorm, or a universal action-hold fix. Full evidence and other groups: docs/VALUE_COMMITMENT_RESULTS_V1.md.

Values on actual searched paths: a selective causal effect

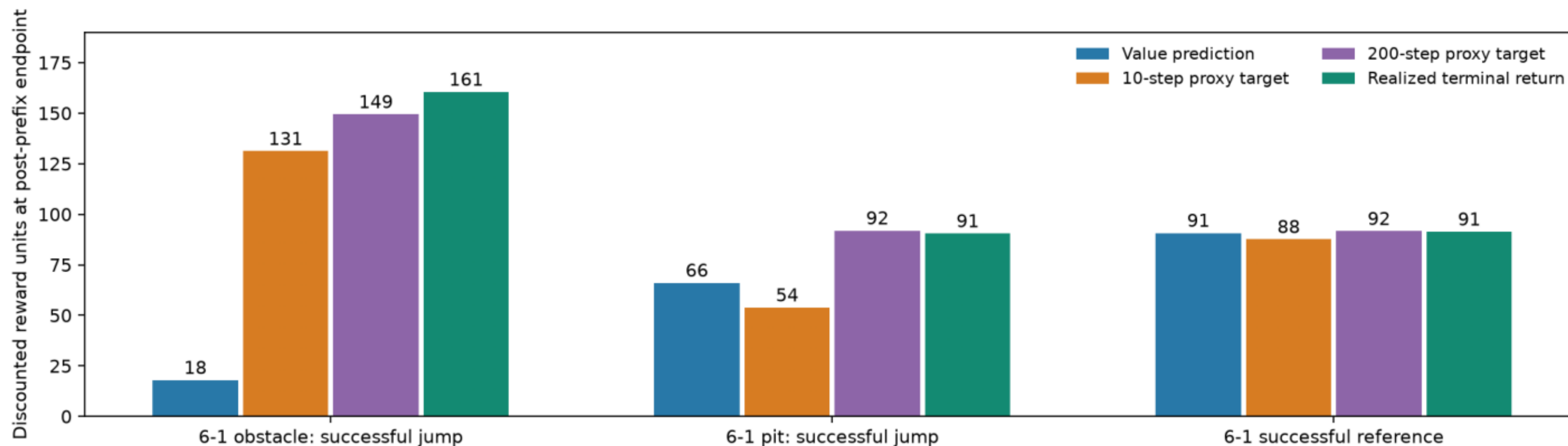
Replacing searched leaf values helps some failures and harms some successes



1,120 new branches; 56 selected roots from 28 source episodes. Repeated branch seeds are conditional, not independent starts.
Privileged emulator access; replacement values are predictions from the same network, not ground truth.
No terminal paths were reached in any arm. Intervention affects subsequent search allocation as well as value backups.

What the completed value-target audit adds

A short value target is not the only possible bottleneck



Illustrative pre-existing roots 40, 35 and 37; position 8; median of 10 conditional continuations per root/arm.
The current online network substitutes for the unavailable historical lagged target network. These are not original learner targets.
At the obstacle root, the 10-step proxy already demands a correction (18 → 131). This does not identify why training missed it.

The next work should answer three decisions

- 1. Breadth: does a fixed gated controller preserve successes and improve completion over all eligible levels?
- 2. Mechanism: do useful states need better replay coverage/fitting, or different value targets?
- 3. Generality: can a student retain specialist skills under the same evaluation protocol?

How someone can pick this up

- Read START_HERE.md, then REPORT.md and the current experiment registry.
- Recreate figures in the pinned lightweight environment; run the bounded compute check for models.
- Use the frozen source bundle and manifests, not a historical default configuration.
- Confirm recipient data/compute access and name a project owner before departure.